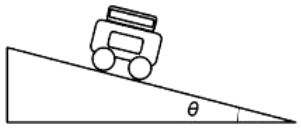
12.2.2 Exercises - Centripetal force 2

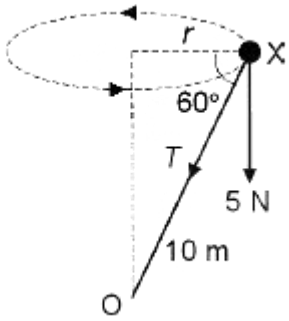
1. The figure below shows a car of mass 1500 kg travels in a horizontal circle of radius 50.0 m along a banked road with a speed of 15 ms^{-1}.

Assuming that the frictional force acting on the car is negligible, calculate the angle θ at which the road is banked.



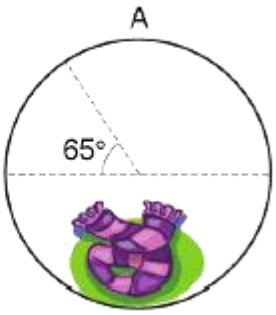
2. A model aeroplane X weight of 5 kg and has a control wire OX of length 10 m attached to it when it flies in a horizontal circle with its wings horizontal, as shown in the diagram below. The wire OX is then inclined at 60° to the horizontal and fixed to a point O and X takes 2s to fly once round its circular path. Assume that g = 10 ms^{-2}.

a. Calculate the tension T in the control wire.



b. Calculate the upward force on X due to the air.

3. In a home laundry dryer, the cylindrical tub of a washer rotates steadily about a horizontal axis. The smooth-walled washer rotates at a steady rate of 0.84 revolutions in one second and the radius of the tub is 0.36 m . Assume that g = 10 ms^{-2}.



a. By considering the forces acting on a piece of cloth, suggest what happens to this piece of cloth as it reaches the highest point, A, in a spin cycle. Support your answer qualitatively.

b. A manufacturer proposed a washer design which he claims is an improvement to the washer design mentioned above. In this 'improved' design, the rate of rotation of the tub is chosen such that a small piece of cloth loses contact with the tub when the cloth is at an angle of 65° with the horizontal, as shown in the diagram above.

i. Calculate the rate of rotation required in this new design.

ii. Suggest two reasons why this washer design is preferred over the previous washer design.

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